

|      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |          |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1(a) | <p><i>structure max 2 marks</i></p> <p><b>1</b> RER has ribosomes <b>R</b> if context is within lumen<br/> <b>or</b><br/> SER does not have ribosomes ;</p> <p><b>2</b> RER, flattened sacs / cisternae, <u>and</u> SER tubular ; AW</p> <p><b>3</b> RER continuous with, (external) nuclear membrane ; <b>ora</b><br/> <b>I</b> near to nuclear membrane<br/> <b>R</b> continuous with <u>internal</u> nuclear membrane</p> <p><b>4</b> <i>ref. to</i> RER, more regular / layered, arrangement<br/> <b>or</b><br/> SER more irregular / disorganised, arrangement ;</p> <p><i>function</i></p> <p><b>5</b> RER produces / transports, proteins / glycoproteins ; <b>A</b> polypeptides<br/> <b>A</b> examples of post-translational modification</p> <p><b>6</b> SER produces, lipids / cholesterol / steroids ; <b>A</b> stores<br/> <b>I</b> produces hormones</p> | <b>3</b> |
| 1(b) | <p><i>any two from</i><br/> <i>ref. to</i> phosphate (group) and two fatty acid, chains / tails / residues ;<br/> <b>A</b> two hydrocarbon chains</p> <p>correct <i>ref. to</i> attachment, phosphate / fatty acids, to glycerol ;</p> <p>AVP ; e.g. <i>ref. to</i> ester bond<br/> <i>ref. to</i> additional group e.g. choline-containing<br/> <i>ref. to</i>, saturated / unsaturated, (fatty acid tails)</p> <p><i>award marks from an annotated diagram</i></p>                                                                                                                                                                                                                                                                                                                                                                                                   | <b>2</b> |

| Question  | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Marks |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 1(c)(i)   | <p>flattened sacs / cisternae ;<br/>stack / layered / described ;</p> <p><i>any one from</i><br/><i>idea that</i> each cisterna in stack is, separate / not interconnected ;<br/>smooth (outer surface) / no ribosomes ;<br/>close to, nucleus / RER ;<br/>single membrane ;<br/>AVP ; e.g. two distinct faces<br/><i>ref. to</i> swollen ends to cisternae</p>                                                                                                                                                                                                                                            | 2     |
| 1(c)(ii)  | <p><i>ignore events occurring before vesicle formation</i></p> <p><i>any three from</i><br/>vesicle moves to, cell surface / cell (surface) membrane ;<br/><b>A</b> plasma membrane</p> <p>movement of vesicle via, cytoskeleton / microtubules ;</p> <p>vesicle fuses with (cell surface) <u>membrane</u> (and protein released) ; AW<br/>e.g. vesicle makes contact with <u>membrane</u> and becomes part of it<br/><b>I</b> attaches to / binds to</p> <p>exocytosis ; <b>I</b> bulk transport / excreted / secreted<br/><i>allow in context of release only or movement from Golgi and release</i></p> | 3     |
| 1(c)(iii) | <p><i>any one relevant e.g.</i><br/>acts as (self-)antigen / recognition site / cellular recognition / receptor / membrane stability / cell adhesion ;<br/><b>I</b> involved in cell signalling</p>                                                                                                                                                                                                                                                                                                                                                                                                        | 1     |